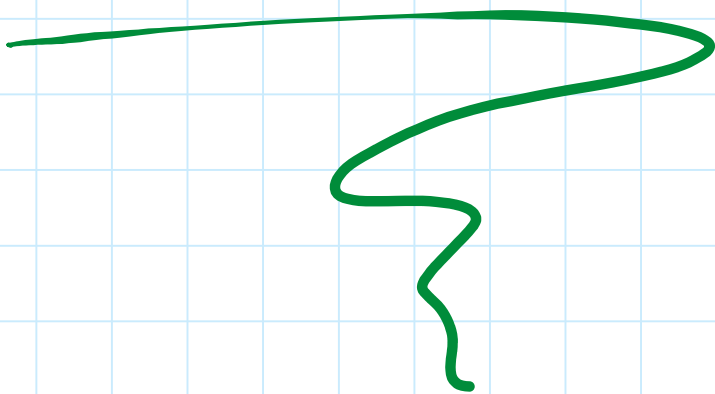


Overview

- 1) Linear algebra → vectors and matrices recap
Systems of equations, matrix inverse
Eigenvalues and eigenvectors.
- 2) Calculus → Ordinary differential equations (ODEs)
Numerical methods
Systems of ODEs.
- 3) Stochastic ODEs → Definition + comparison to ODEs
Stochastic LIF model

After each section, we will do some exercises and go through the jupyter notebooks.

LINEAR ALGEBRA

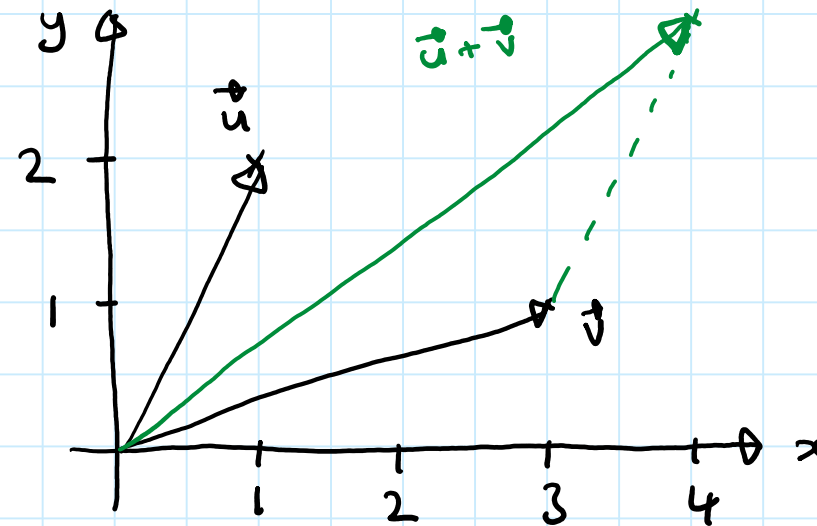


VECTORS AND MATRICES

Geometric view ,

$$\vec{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\vec{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$



Add vectors , $\vec{u} + \vec{v} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$

Dot product , $\vec{u} \cdot \vec{v} = \vec{u}^T \vec{v} = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 3 + 2 = 5$

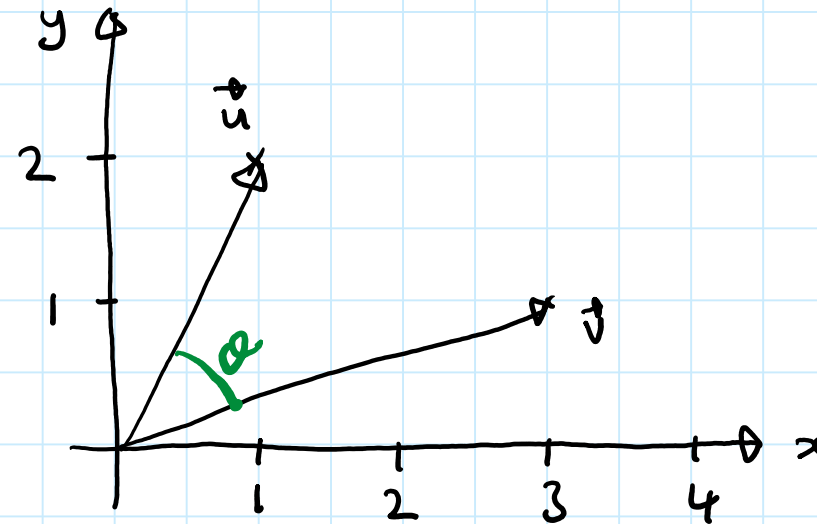
Length of a vector , $|\vec{u}| = \sqrt{\vec{u} \cdot \vec{u}} = \sqrt{1^2 + 2^2} = \sqrt{5}$
 $|\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}} = \sqrt{3^2 + 1^2} = \sqrt{10}$ } Pythagoras theorem

VECTORS AND MATRICES

Geometric view ,

$$\vec{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\vec{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$



angles between vectors, $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$

perpendicular $\Rightarrow \vec{a} \cdot \vec{b} = 0$, e.g.) $\vec{a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow$ vectors are at right angles

we define unit vectors as having length 1 , e.g.) $\hat{u} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

We can use matrices to transform vectors.

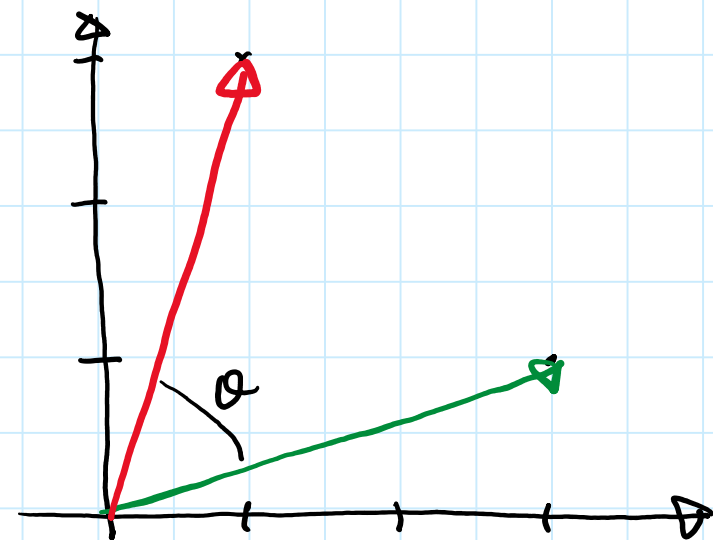
Say we have, $\vec{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, we can transform it like

$$\vec{v}' = A \vec{v}, \quad A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\vec{v}' = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Identity matrix: $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $I \vec{u} = \vec{u}$
 (does nothing but shows up everywhere)

Rotation matrix: $R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$



$$\vec{u}' = R(\theta) \vec{u}, \quad \text{preserves lengths, } |\vec{u}'| = |\vec{u}|.$$

SYSTEMS OF EQUATIONS

Say we have ,
$$\begin{aligned} 2x + y &= 1 \\ 5x + y &= 4 \end{aligned} \rightarrow \begin{pmatrix} 2 & 1 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$A \vec{u} = \vec{v}$$

Solve with the matrix inverse.

The matrix inverse is defined s.t. $A^{-1} A = \mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

If we know the inverse, then:

$$A \vec{u} = \vec{v}$$

$$\underbrace{A^{-1} A}_{\mathbb{I}} \vec{u} = A^{-1} \vec{v}$$
$$\vec{u} = A^{-1} \vec{v}$$

SYSTEMS OF EQUATIONS

Calculating the inverse, for a 2D matrix.

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det(M) = ad - bc$$

(determinant)

$$\rightarrow M^{-1} = \frac{1}{\det(M)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad \rightarrow \text{check } M^{-1}M = I.$$

For $A = \begin{pmatrix} 2 & 1 \\ 5 & 4 \end{pmatrix}$, $\det(A) = 8 - 5 = 3$, $A^{-1} = \frac{1}{3} \begin{pmatrix} 4 & -1 \\ -5 & 2 \end{pmatrix}$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 4 & -1 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 0 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \rightarrow x=0, y=1.$$

If $\det(M) = 0 \Rightarrow M$ has no inverse.

EIGENVALUES & EIGENVECTORS

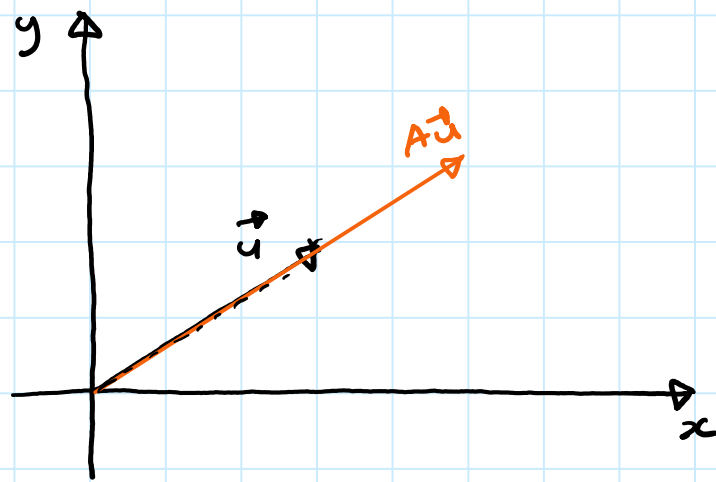
Eigenvalue equations

$$A \vec{u} = \lambda \vec{u}$$

matrix $\nearrow$
 $\nwarrow$ eigenvalue
 $\nwarrow$ eigenvectors $\nearrow$

$$\left[\begin{array}{l} \text{before, we had} \\ A \vec{u} = \vec{v} \end{array} \right]$$

The transformation A doesn't rotate the eigenvectors, it changes their length by a factor of λ , the eigenvalue



A 2×2 matrix can have 2 eigenvalues and eigenvectors.

But not all matrices have them.

Also means that

$$A A \vec{u} = \lambda^2 \vec{u}$$

$$A^3 \vec{u} = \lambda^3 \vec{u}$$

$$A^n \vec{u} = \lambda^n \vec{u}$$

$$A \vec{u} = \lambda \vec{u} \quad \longrightarrow \quad (A - \lambda \mathbb{I}) \vec{u} = 0$$
$$\vec{u} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \det(A - \lambda I) = 0$$

Say $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $A - \lambda \underline{1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix}$

→ solve to find λ_1 and λ_2

vectors and matrices Page 9

Example

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}, \quad A - \lambda \mathbb{1} = \begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$$

$$\det(A - \lambda \mathbb{1}) = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 \\ = (\lambda - 2)(\lambda - 5)$$

$$\rightarrow \lambda_1 = 2, \quad \lambda_2 = 5$$

eigenvectors: $\lambda_1: (A - \lambda_1 \mathbb{1}) \vec{u} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$2x + y = 0$$

$$\lambda_1 = 2, \quad u_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\lambda_2: (A - \lambda_2 \mathbb{1}) \vec{u} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$x - y = 0$$

$$\lambda_2 = 5, \quad u_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

DIAGONALIZATION

In general, say A is an $n \times n$ matrix $\rightarrow$ has n eigenvalues & eigenvectors.

Define a matrix $X = (\vec{u}_1 \ \vec{u}_2 \ \dots \ \vec{u}_n)$

$\uparrow$
each column is an eigenvector.

$$\Rightarrow AX = (\lambda_1 \vec{u}_1 \ \lambda_2 \vec{u}_2 \ \dots \ \lambda_n \vec{u}_n) \\ = X\Lambda, \quad \Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{pmatrix}$$

With the inverse of X , we get $X^{-1}AX = \Lambda$ (*)

We say that X diagonalises A .

$\rightarrow$ Also $A = X\Lambda X^{-1}$, $A^2 = X\underline{\Lambda X^{-1}X} \Lambda X^{-1} = X\Lambda^2 X^{-1}$

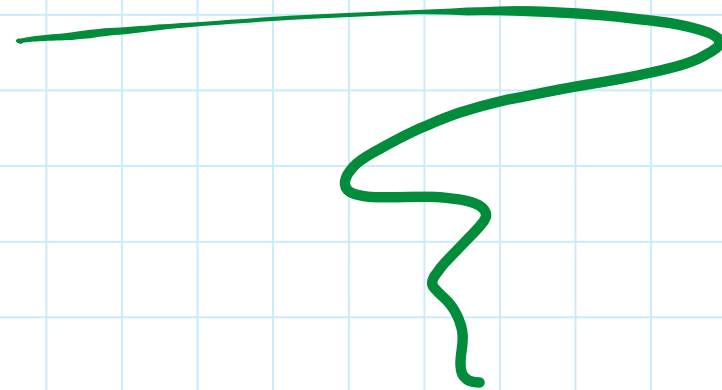
$$\dots A^k = X\Lambda^k X^{-1}, \quad \Lambda^k = \begin{pmatrix} \lambda_1^k & & 0 \\ & \lambda_2^k & \\ 0 & & \ddots \\ & & & \lambda_n^k \end{pmatrix}$$

QUESTIONS

- 1) For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, check that $M^{-1} M = \mathbb{1}$.
 - 2) Solve $2x + y = 5$, $x - y = 1$
 - 3) Solve $5x + 2y = 14$, $3x - 4y = -2$
 - 4) Solve $-2x + y = 3$, $4x + 5y = -6$
 - 5) Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 6 & -2 \\ -2 & 3 \end{pmatrix}$
- } using matrix inverses

Plot the eigenvectors, and show how transforming them by A changes them.

CALCULUS



ORDINARY DIFFERENTIAL EQUATIONS

DEs tell us how quantities change with time or position.

Eg) If we have a function $y(t)$, $\frac{dy}{dt} = \frac{y(t+dt) - y(t)}{dt} \rightarrow$ rate of change of $y(t)$.

$\rightarrow$ the DE: $\frac{dy}{dt} = f(y, t)$

this function tells us
how $y(t)$ changes in time.

Goal: find the solution $y(t)$ exactly

this is not always possible $\rightarrow$ numerical approximations to the solution

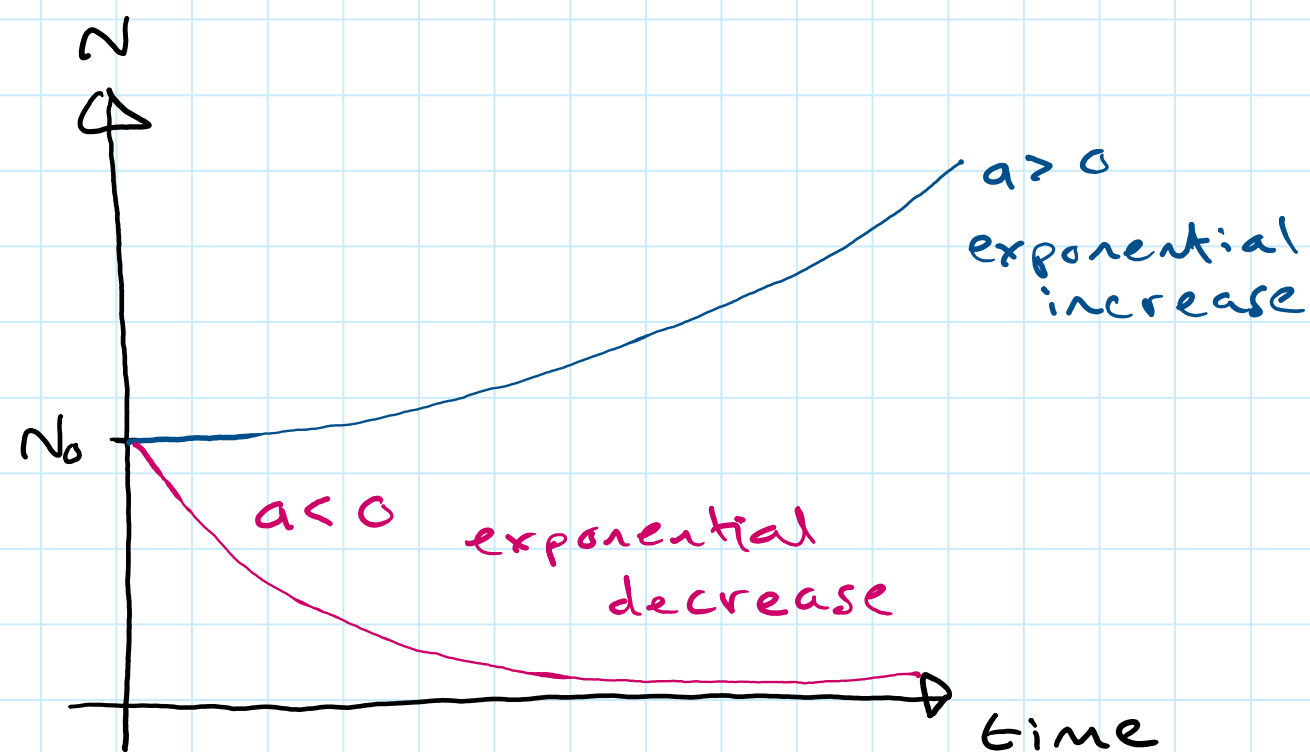
$\rightarrow$ 'simulations'

Exact solution examples

We have some population $N(t)$ that changes in time.

Described by: $\frac{dN}{dt} = aN$

We can guess the solution because we know how exponentials work.



$$\frac{1}{aN} dN = dt$$

$$\int \frac{1}{aN} dN = \int dt$$

$$\frac{1}{a} \ln(N) = t + C$$

$$N = N_0 e^{at}$$

↑
population at $t=0$.

this example is linear — no N^2 terms or higher

and first-order — no $\frac{d^2 N}{dt^2}$ terms or higher

Exact solution examples

At second order we can get periodic behaviour.

$$y(t): \quad \frac{d^2 y}{dt^2} = a y$$

if $a > 0$, we have another exponential solution:

$$y(t) = A e^{\sqrt{a}t} + B e^{-\sqrt{a}t}$$

if $a < 0$, same solution but we have $\sqrt{\text{negative number}}$.

→ imaginary numbers.

we define $i = \sqrt{-1}$ as the imaginary unit, $i^2 = -1$

Complex numbers have a real and imaginary component.

$$z = a + ib \quad a \text{ and } b \text{ are real.}$$

Exact solution examples

At second order we can get periodic behaviour.

Rewrite: $\frac{d^2 y}{dt^2} = a y \rightarrow \frac{d^2 y}{dt^2} = -a^2 y$, a is real

Then the solution is:

$$y(t) = A e^{iat} + B e^{-iat}$$

Euler's theorem: $e^{i\theta} = \cos(\theta) + i \sin(\theta) \rightarrow$ complex number

We always take the real part of the solution at the end.

$\operatorname{Re}[e^{i\theta}] = \cos\theta$, $\operatorname{Im}[e^{i\theta}] = \sin\theta \rightarrow$ check differentiation rules.

Exact solution examples → Leaky integrate & fire model

LiF is a model for how neurons spike when a current is applied.

$V(t)$ is the membrane potential.

$$C_m \frac{dV(t)}{dt} = - \underbrace{g_L(V(t) - E_L)}_{\text{leak current}} + I(t)$$

capacitance

input current

The potential $V(t)$ evolves in time

when $V(t) > V_{\text{threshold}}$, neuron spikes and $V(t) \xrightarrow{\text{resets}} V_{\text{reset}}$.
(fires)

Exact solution examples $\rightarrow$ Leaky integrate & fire model

LIF is a model for how neurons spike when a current is applied.

$$C_m \frac{dV(t)}{dt} = -g_L (V(t) - E_L) + I(t)$$

To solve, assume a constant input potential, $I(t) = I_0$, and rewrite:

$$\frac{dV}{dt} + \frac{g_L}{C_m} V = \frac{g_L}{C_m} E_L + \frac{1}{C_m} I_0$$

$$\frac{d}{dt} \left(e^{\frac{g_L}{C_m} t} V \right) = \left(\frac{g_L}{C_m} E_L + \frac{1}{C_m} I_0 \right) e^{\frac{g_L}{C_m} t}$$

Now integrate:

$$\begin{aligned} V(t) &= E_L + \tau_m R_m I_0 + C e^{-t/\tau_m}, \quad \tau_m = C_m / g_L \\ &= V_\infty + (V_0 - V_\infty) e^{-t/\tau_m} \end{aligned}$$

$R_m = 1/g_L$

NUMERICAL METHODSEuler's method

First order ODE $\rightarrow \frac{dy}{dt} = f(y, t)$

In infinitesimal form, we write $\frac{dy}{dt} = \frac{y(t+dt) - y(t)}{dt}$

dt is a very small time-step

So we can re-write the ODE as $y(t+dt) = y(t) + f(y, t)dt$

$\Rightarrow$ we can evolve the solution from an initial time t_0 .

1) Choose t_0 , $y(t_0)$, and dt .

2) $y(t_0 + dt) = y(t_0) + f(y, t_0)dt$

3) $y(t_0 + 2dt) = y(t_0 + dt) + f(y, t_0 + dt)dt$

4) $y(t_0 + 3dt) = y(t_0 + 2dt) + f(y, t_0 + 2dt)dt$

⋮ ⋮ ⋮

$y(t + Ndt)$

NUMERICAL METHODSRUNGE-KUTTA METHOD

We can Taylor expand the solution about $t = t_0$

$$y(t) = y(t_0) + dt f(y, t_0) + \frac{1}{2!} dt^2 \frac{d}{dt} F(y, t_0) + \dots \quad dt = t - t_0$$

Euler's method

Euler's method

$O(dt^4)$

Runge-Kutta method

The update rule is:

$$y(t+dt) = y(t) + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = dt f(y, t)$$

$$k_2 = dt f(y + \frac{1}{2}k_1, t + \frac{1}{2}dt)$$

$$k_3 = dt f(y + \frac{1}{2}k_2, t + \frac{1}{2}dt)$$

$$k_4 = dt f(y + dt k_3, t + dt)$$

SYSTEMS OF ODEs

Say we have two coupled ODEs :

$$\frac{dx}{dt} - y = 0$$

$$\frac{dy}{dt} + 2x + 3y = 0$$

We can write this as a matrix equation

$$\frac{d\vec{u}}{dt} = A \vec{u}$$

where $\vec{u} = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$, and $A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$

If we write $\vec{u} = e^{\lambda_1 t} \vec{v}_1$, where $\vec{v}_1$ is constant in time we get

$$\frac{d\vec{u}}{dt} = \lambda_1 \vec{u} \Rightarrow A \vec{v}_1 = \lambda_1 \vec{v}_1 \rightarrow \text{eigenvalue equation}$$

A 2×2 matrix has 2 eigenvalues and eigenvectors

$\rightarrow$ the general solution is : $\vec{u}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{\lambda_1 t} \vec{v}_1 + e^{\lambda_2 t} \vec{v}_2$

SYSTEMS OF ODEs

Say we have two coupled ODEs :

$$\frac{dx}{dt} - y = 0$$

$$\frac{dy}{dt} + 2x + 3y = 0$$

We can write this as a matrix equation

$$\frac{d\vec{u}}{dt} = A \vec{u}$$

where $\vec{u} = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$, and $A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$

The solution : $\vec{u}(t) = c_1 e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$

integration constants

$\lambda_1 = -1$
 $\lambda_2 = -2$

$\vec{v}_1$ $\vec{v}_2$

Asymptotic behaviour : As $t \rightarrow \infty$, $\vec{u} \rightarrow c_1 \vec{v}_1 + c_2 \vec{v}_2$

$$\frac{dx}{dt} = 0 \quad \& \quad \frac{dy}{dt} = 0 \quad \rightarrow \text{equilibrium points.}$$

SYSTEMS OF ODEs

Take the general case: $\frac{dx}{dt} = f(x, y)$, $\frac{dy}{dt} = g(x, y)$
(possibly non-linear)

Say we have an equilibrium point at $(\bar{x}, \bar{y}) \Rightarrow \frac{d\bar{x}}{dt} = \frac{d\bar{y}}{dt} = 0$

Goal: study / characterize it.

$\rightarrow$ study small perturbations around $(\bar{x}, \bar{y})$

Step 1

Taylor expand around $(\bar{x}, \bar{y})$, $x = \bar{x} + p$, $y = \bar{y} + q$

$$\frac{d(\bar{x} + p)}{dt} = \frac{dp}{dt} = f(\bar{x}, \bar{y}) + \frac{\partial f}{\partial x} p + \frac{\partial f}{\partial y} q$$

$$\frac{d(\bar{y} + q)}{dt} = \frac{dq}{dt} = g(\bar{x}, \bar{y}) + \frac{\partial g}{\partial x} p + \frac{\partial g}{\partial y} q$$

SYSTEMS OF ODEs

Take the general case: $\frac{dx}{dt} = f(x, y)$, $\frac{dy}{dt} = g(x, y)$

Step 2 write the DE's for the perturbations p and q as a matrix equation

$$\vec{u} = \begin{pmatrix} p \\ q \end{pmatrix}, \quad \frac{d\vec{u}}{dt} = J \vec{u}, \quad J = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} \quad \text{The Jacobian}$$

Step 3 Find the solutions, i.e. eigenvalues & eigenvectors of J .

$$\vec{u} = c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2$$

λ_1 and $\lambda_2 < 0$ $\rightarrow$ stable, small perturbations near equilibrium fade away in time.

$\lambda_1 < 0$ and $\lambda_2 > 0$ $\rightarrow$ unstable saddle point

λ_1 and $\lambda_2 > 0$ $\rightarrow$ unstable

SYSTEMS OF ODEs

So far, we have considered changes with time.

But in a physical system, other things can change as well.

For example, in the LIF model, the current I_0 can change.

→ this can cause changes in the type and number of equilibrium points.

We call this a bifurcation.

Eg 1) Saddle-node bifurcation → two equilibria are created or destroyed as a parameter in the system is changed.

Eg 2) Hopf - bifurcation → the eigenvalues of J become complex as a parameter in the system is changed.
⇒ periodic behaviour in the growth of the perturbations.

QUESTIONS

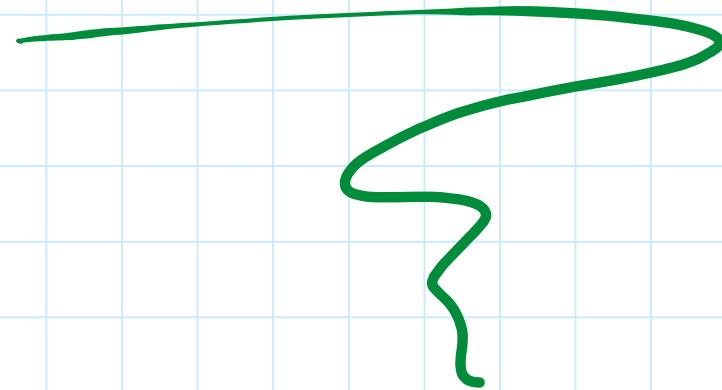
- 1) Calculate $\sqrt{i}$
- 2) Write $\cos(3x)$ and $\sin(3x)$ in terms of exponentials
- 3) Solve $\frac{dy}{dt} - 4y = 0$, $y(0) = 3$
- 4) Solve $\frac{dy}{dt} - 2y = e^t$, $y(0) = 0$
- 5) Using the LIF solution $V(t)$, derive the time-to-spike relation for a threshold potential $V_{\text{threshold}}$. (calculate how long the neuron takes to spike)
- 6) Calculate the Runge-Kutta update for exponential growth, $\frac{dN}{dt} = aN$.
→ then, verify that it is accurate to 4th order in the Taylor expansion.
- 7) Solve this system of ODEs using matrix methods:
$$\begin{aligned}\frac{dx}{dt} &= y \\ \frac{dy}{dt} &= -6x - 5y\end{aligned}$$

next page →

QUESTIONS

- 8) nonlinear case: $\frac{dx}{dt} = \mu - x^2$ $\rightarrow$ a) find all equilibria as a function of μ
 $\frac{dy}{dt} = -y$ b) compute the Jacobian + find eigenvalues
c) classify the equilibria
d) discuss bifurcation in the system.

STOCHASTIC DIFFERENTIAL EQUATIONS



ODEs

$$\frac{dy}{dt} = f(y, t)$$

$$dy = f(y, t) dt$$

$$y(t+dt) = y(t) + f(y, t) dt$$

Then we can simulate:

$$y(0) \rightarrow y(dt) \rightarrow y(2dt) \rightarrow \dots$$

$f(y, 0) dt$ $f(y, dt) dt$

Stochastic DEs (SDEs)

$$dy = f(y, t) dt + g(y, t) \underbrace{dW(t)}_{\downarrow}$$

$dW(t)$ is a random noise contribution to the equation

→ Brownian motion, or, Wiener process

$$\text{So, } y(t+dt) = y(t) + f(y, t) dt$$

f : drift term

g : diffusion term

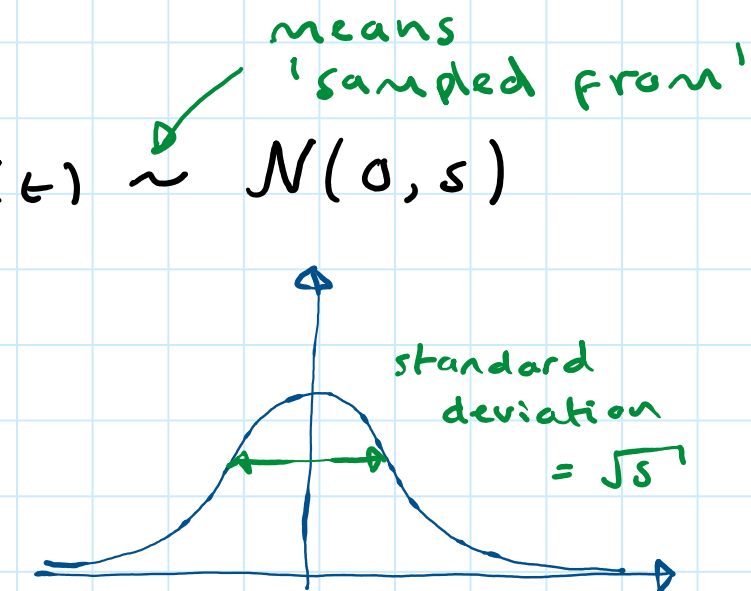
$$+ g(y, t) dW(t)$$

a random kick to the trajectory at each time step.

BROWNIAN MOTION

What is $w(t)$? A real-valued, continuous-time stochastic process.

- At $t=0$, $w(0) = 0$
- $w(t)$ has Gaussian increments: $w(t+s) - w(t) \sim \mathcal{N}(0, s)$
- $w(t)$ is the accumulation of noise from $t=0$ to some time $t>0$.
- Note: $w(t)$ is not a distribution
At every t , $w(t)$ is a single number
- We're doing calculus, so we need the infinitesimal increment of $w(t)$:



$$dw(t) = w(t+dt) - w(t) \sim \mathcal{N}(0, dt)$$

$$\int_0^t dw(r) = w(t)$$

BROWNIAN MOTION

A simple example, the SDE for Brownian motion:

$$dX_t = \mu dt + \sigma dW_t$$

we often use the subscript dX_t instead of $dX(t)$ with SDEs.

μ : a constant drift term

σ : a constant noise coefficient

To solve, simply integrate both sides:

$$\int dX_t = \mu \int dt + \sigma \int dW_t$$

$$\rightarrow X_t = x_0 + \mu t + \sigma W_t$$

$\uparrow$
integration
constant $\rightarrow$ starting
position

1) particle starts at x_0

2) it moves with velocity μ

3) there is a noise contribution to the motion, tracked by σW_t .

EULER - MARUYAMA

simulating SDEs

A general SDE looks like: $dX_t = a(x_t, t)dt + b(x_t, t)dW_t$

↑ generic functions for
drift and noise

- Like the Euler method, we write

$$dX_t = X_{t+1} - X_t$$

- Then the update rule is:

$$X_{t+1} = X_t + a(x_t, t)dt + b(x_t, t)dW_t$$

So we start at $X_t = X_0$, choose a dt , and generate/simulate the solution.

- The noise term $dW_t \sim \mathcal{N}(0, dt) \equiv dW_t \sim \sqrt{dt} \mathcal{N}(0, 1)$

→ the update: $X_{t+1} = X_t + a(x_t, t)dt + b(x_t, t)\sqrt{dt}\xi_0$

where $\xi_0 \sim \mathcal{N}(0, 1)$.

STOCHASTIC CALCULUS

In calculus, the chain rule let's us change variables.

Eg if we have $\frac{dx}{dt} = g(x, t)$, and change variables $y = f(x(t))$

chain rule: $\frac{dx}{dt} = \left(\frac{df}{dx}\right)^{-1} \frac{dy}{dt} \rightarrow \frac{dy}{dt} = \left(\frac{df}{dx}\right) g(y, t)$

g is re-written
in terms of $y(t)$

Stochastic calculus is different.

Starting with $dx_t = a(x_t, t) dt + b(x_t, t) dW_t$

if we change variables: $y_t = f(x_t)$

we get: $dy_t = \left(a(x_t, t) \frac{df}{dx_t} + \underbrace{\frac{1}{2} b(x_t, t) \frac{d^2 f}{dx^2}}_{\text{this term is new}} \right) dt + \frac{df}{dx_t} b(x_t, t) dW_t$

this term is new

it arises because $dW_t \sim \sqrt{t}$

this is called Ito's calculus.

QUESTIONS

1) The SDE for geometric Brownian motion is :

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

Change variables, re-write the SDE in terms of $Y_t = \ln X_t$.

2) Solve the SDE, and put the solution back in terms of X_t .